

Web-Assotest: a browser-based tool for genetic association analysis under multiple inheritance models

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Summary

Web-Assotest is a browser-based tool for analysing the association between a genetic variant and a binary outcome (cases vs. controls) from a 3x2 table of observed genotype counts (wild-type / heterozygote / homozygote, by case/control status).

The tool simultaneously fits a hierarchy of five nested logistic-regression models — null, dominant, co-dominant (additive), recessive, and genotypic — and presents the resulting odds ratios, 95% confidence intervals, likelihood-ratio test statistics, and AIC values in a single model-comparison diagram, together with a Hardy-Weinberg equilibrium (HWE) check for cases, controls, and the combined sample (Wigginton, Cutler, and Abecasis 2005). Users can optionally fit the same model hierarchy with Firth’s penalized logistic regression (Firth 1993; Ploner et al. 2023) to obtain stable estimates when a genotype count is zero or the data are otherwise separated.

Web-Assotest is implemented in R as a single `flexdashboard`/Shiny application (R Core Team 2024; Chang et al. 2024; Iannone, Allaire, and Schloerke 2024), runs entirely in the browser once deployed, and requires no local software installation, scripting, or statistical training beyond entering the genotype counts into an editable table.

Non-R-savvy users who do not want to run the application locally can run the application directly in a browser from this URL <http://ekstroem.com:3838/webassotest/>.

Statement of need

Testing whether a genetic variant is associated with a disease or trait from a simple genotype-by-outcome count table is one of the most common analyses in applied genetic epidemiology, and it is one that many applied researchers (clinicians, wet-lab biologists, or students) need to perform without necessarily writing R or Python code themselves.

The statistical subtlety is that “the” genetic effect is not a single quantity: the same data can be tested under a dominant, recessive, co-dominant, or fully genotypic model, and these choices are not interchangeable. A variant that shows a strong dominant effect can show a null recessive effect, and vice versa.

Existing options for this analysis are either large, general-purpose genetics toolchains aimed at genome-wide data (e.g. PLINK) that are overkill and require command-line fluency for a single 3x2 table, or R packages such as `SNPassoc` that require the user to already be working in R. **Web-Assotest** fills the gap between these: a lightweight, model-comprehensive, point-and-click tool for the single-marker case, so that the choice of genetic model is made visible and testable rather than assumed. The exact HWE test of Wigginton, Cutler, and Abecasis (2005) is bundled so that genotyping-quality checks and the association test itself are available in one place. The tool has been used as teaching material for case-control study design and has been applied in the author’s own applied genetics research (see Research impact statement below).

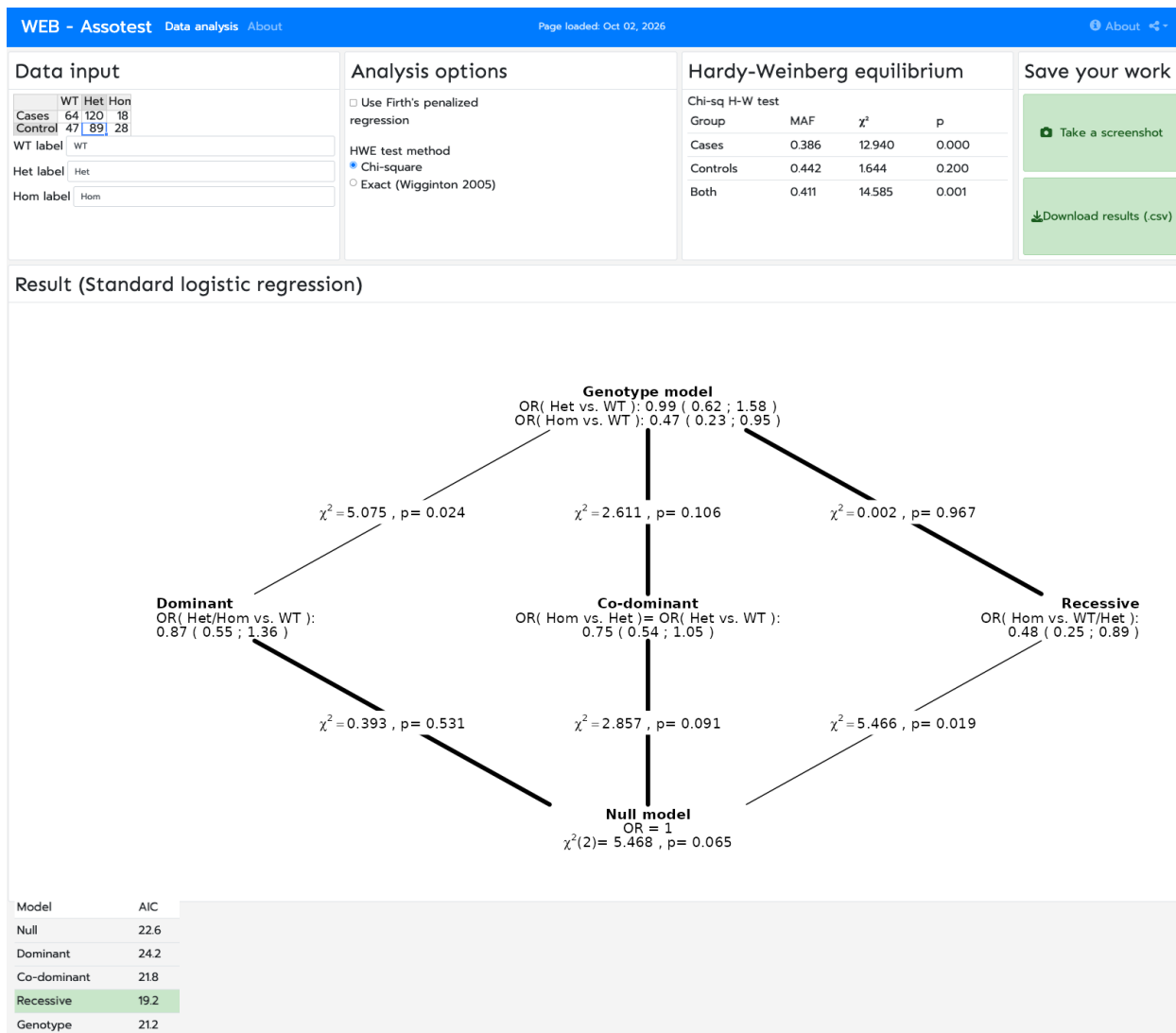


Figure 1: The web-interface for Web-Assotest.

State of the field

TODO: expand with a short, referenced comparison to alternative tools users might reach for instead of `Web-Assotest`, for example:

- Command-line/whole-genome toolchains (e.g. PLINK) — powerful for genome-wide data but require installation and scripting for a single marker.
- R packages for candidate-gene association (e.g. `SNPassoc`, `genetics`) — flexible but require R programming and are not accessible to non-programming collaborators.
- Standalone HWE calculators (e.g. the Court Lab HWE calculator) — cover HWE testing only, not the genetic-model comparison this tool provides.

`Web-Assotest` differs from all of the above by combining (a) a no-install, browser-based interface, (b) simultaneous fitting and visual comparison of all standard genetic models rather than a single pre-chosen contrast, and (c) an integrated HWE check, in one small single-file application.

Software design

`Web-Assotest` is implemented as a single R Markdown file (`webassotest.Rmd`) built with `flexdashboard` and running as a Shiny application (`runtime: shiny`), so there is no separate `ui.R/server.R` split.

The reactive pipeline reads the user's edits to a 3×2 `rhandsontable` genotype-count table, rounds and validates them, and re-fits all five nested models (`nul`, `gen`, `dom`, `cod`, `rec`) on every change via the function `gtm()`. The application then draws the model-comparison diagram with base R graphics and returns a tidy summary data frame used for both the on-screen Akaike information criterion (AIC) table and CSV export. Likelihood-ratio tests use `anova(..., test = "Chisq")` under standard logistic regression, or a manual deviance difference under Firth's penalized regression (Firth 1993) via the `logistf` package (Ploner et al. 2023); the latter is offered as a toggle so that zero-cell/separated genotype counts do not require the Fisher's-exact-test fallback used in the standard mode. HWE testing (`SNPHWE()`) implements the exact test of Wigginton, Cutler, and Abecasis (2005) directly in R rather than depending on an external genetics package, keeping the application to a single deployable file with a small dependency footprint (`flexdashboard`, `rhandsontable`, `shinyscreenshot`, `logistf`).

Research impact statement

`Web-Assotest` (and its predecessor) has been used in published candidate-gene case-control studies spanning more than two decades (2004–2025) and a broad range of research groups and populations, including Danish, Polish, Iranian, Japanese, Iraqi, and Indian cohorts. The applications cover a wide range of phenotypes and publications, some of which are listed below:

- **Type 2 diabetes and its complications**, including diabetic retinopathy and diabetic foot (S. K. Hansen et al. 2004; S. Hansen et al. 2005; Keshavarz et al. 2006; Mrozikiewicz-Rakowska et al. 2017; Gohari-Lasaki et al. 2020).
- **Obesity, adipokines, and ageing** (Roszkowska-Gancarz et al. 2012; Al-Ogaidi, Abdulsattar, and Al-Dulaimi 2019).
- **Cardiovascular disease**, including essential hypertension and acute coronary syndrome (Wrzosek et al. 2015; Gutowska et al. 2025).
- **Psychiatric and behavioural phenotypes**, including completed suicide, opioid dependence, and paraphilic sexual offending (Chojnicka et al. 2012, 2013, 2014; Fudalej et al. 2016; Jakubczyk et al. 2017).
- **Neurodegenerative disease** (late-onset Alzheimer's disease) (Fatemi Langroudi, Zeinaly, and Ajamian 2023).
- **Autoimmune disease** (Graves' disease) (Kurylowicz et al. 2007).
- **Ophthalmology** (Fuchs endothelial corneal dystrophy) (Ołdak et al. 2015).
- **Gastrointestinal disease** (colonic diverticulosis) (Nehring et al. 2023).

- **Reproductive health** (recurrent miscarriage) (Sudhir et al. 2016).

This breadth illustrates that the tool meets a recurring need in applied genetic epidemiology across clinical specialties: a quick, transparent comparison of genetic inheritance models for a single marker, performed by researchers who are not necessarily statistical programmers.

AI usage disclosure

AI has been used to correct this manuscript for spelling and Claude Code was used to assemble the initial list of papers citing Web-Assotest. The list has subsequently been manually curated. The core part of the `webassotest.Rmd` application predates AI tools and has not been. Claude Code has been used to update and improve the flexdashboard layout and has been used to optimize the load speed of the reactive expressions in the application.

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